

Keshav Krishna

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EDUCATION

New York University Courant

MS, Computer Science (GPA: 4.0)

- **Coursework:** Machine Learning, Honors Analysis of Algorithms, Programming Languages

Aug 2025 - May 2027

New York City

Indian Institute of Technology, Ropar

Bachelors, Computer Science (GPA: 8.74/10)

- **Achievements:** Merit Scholarship for top performance

Aug 2019 - Jun 2023

Ropar, India

SKILLS SUMMARY

- **Programming Languages & Concepts:** C, C++, Python, SQL, Unix scripting, Data Structures
- **Data Engineering & Storage:** Postgres, MYSQL, redis, hive, pyspark, Airflow, Hadoop, Apache Spark, Apache Flink, NoSQL, ETL, databases, JSON
- **Machine Learning & AI:** Machine learning, deep learning, LLMs, Tensorflow, Hugging Face, Pytorch, Scikit-learn, Machine Learning Frameworks
- **DevOps & Orchestration:** Docker, GIT, JIRA, Kubernetes, Orchestration

EXPERIENCE

JioSaavn | Software Engineer(Data Science)

Jul 2023 - Jul 2025

- Built early-morning and late-night content mixes for 12M daily users using Hive, Apache Spark for distributed data processing, Redis, and PHP, driving 1M impressions, 800K clicks, and 500K streams per day
- Generated mood-based topic mixes for 7.6M daily users using Hive, Apache Spark, and scikit-learn's TF-IDF, added deduplication filters and Flask endpoints to the PHP backend, and served them to 100M app users, achieving a 12.81% CTR and 8.5 streams per user
- Assessed few-shot SetFit and transformer models (hing-roberta, muril-base) using machine learning frameworks and transitioned to prompt engineering with Llama3-8B-instruct, achieving 85% accuracy on manually labeled data, thus improving the content moderation of user-generated playlists
- Created a custom BERT model to generate song and user embeddings, which raised top-5 and top-10 accuracy to 61.82% and 67.21%, thereby improving user-learning capabilities and recommendation personalization
- Developed a RAG system using LLaMA3-7B for prompt intent classification, dynamically retrieving songs to assemble playlists and generating custom cover art via the segmid/SSD-1B diffusion model, containerized and deployed with Kubernetes to enhance user interaction and playlist customization
- Reduced infrastructure costs by migrating data from Annoy indices to Qdrant, purging unused data, and cleaning tables for improved lifecycle management

GE Healthcare | Software Intern

May 2022 - Jul 2022

- Built efficient data structures in the image preprocessing pipeline and applied the YOLO algorithm to train a deep-learning model that identified and blurred faces in medical images, achieving over 90% accuracy on more than 20,000 images
- Trained YOLO models to detect people in images(IR and RGB), reaching over 85% mean average precision and accuracy
- Created Docker images with globally accessible modules, simplifying import and utilization in various projects by implementing FastAPI endpoints

RESEARCH & ACADEMIC PROJECTS

Review of machine learning innovations in cache management

Jan 2024 - Apr 2024

<https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2025.1441250/abstract>

Independent Research

- Single-author literature review of 40+ papers on ML for cache systems (replacement policies, dynamic management, edge/content caching, security).
- Evaluated methods (reinforcement learning, LSTM, imitation learning, deep neural nets) for performance optimization and attack detection; synthesized gaps and practical recommendations.
- Published in Frontiers in AI; provided roadmap for research and practitioner adoption with 9+ citations in few months.

FlashAttention-2: From-Scratch Triton Implementation with GPU Profiling & Benchmarking

Jan 2026 - Mar 2026

<https://github.com/k-keshav-k/Flash-Attention>

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- Implemented FlashAttention-2 forward pass in Python as a custom Triton GPU kernel with online softmax and tiled computation, reducing attention memory from $O(N^2)$ to $O(N)$ and enabling sequence lengths up to 65K
- Built backward pass using activation recomputation from stored logsumexp values, compiled with for kernel fusion across FP32 and BF16 precisions
- Profiled Transformer models up to 2.7B parameters using NVIDIA Nsight Systems and PyTorch memory snapshots within a Docker container, and benchmarked the Triton kernel against vanilla PyTorch and torch.compile baselines across sequence lengths, dimensions, and precisions

Transformer Language Model from Scratch: BPE Tokenizer, Training, and Architecture Ablations

Jan 2026 - Mar 2026

<https://github.com/k-keshav-k/Large-Language-Models-from-Scratch/>

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- Built a complete Transformer language model from scratch in PyTorch (no nn.Linear, nn.Embedding, nn.functional, or torch.optim), implementing BPE tokenizer, RoPE, RMSNorm, SwiGLU, causal multi-head attention, AdamW, cosine LR scheduling, and gradient clipping
- Trained a 17M-parameter pre-norm Transformer on TinyStories (~328M tokens) and ran systematic ablations - RMSNorm removal, pre-norm vs. post-norm, RoPE vs. NoPE, SwiGLU vs. SiLU - with automated hyperparameter sweeps over learning rates and batch sizes
- Performed FLOPs and memory accounting across GPT-2 configurations (124M-1.5B), analysing how attention vs. FFN compute shifts with model scale and context length